112112: K-feldspar volcanic rock, Jump Up Dam

Location and sampling

EDJUDINA (SH 51-6) AMG Zone 51, 412300E 6710100N
Sampled on 13 November 1991

Sampling locality is about 2 km southeast of Jump Up Dam.

Tectonic unit/relations

This sample is of a felsic volcanic unit within the Kurnalpi Domain of the Kurnalpi Terrane. The sample is a fine- to medium grained, banded, quartzofeldspathic rock, possibly a felsic volcanic unit, from a 20 m-thick horizon interleaved with tholeiitic basalt.

Petrographic description

This is a carbonated porphyritic rhyodacite with a remarkably large amount of K-feldspar. The apparent proportion of K-feldspar is probably enhanced by saussuritization and carbonatization of plagioclase, and the dismemberment of plagioclase during late-magmatic flow. Phenocrysts of all three major minerals — quartz, plagioclase, and K-feldspar — are abundant. Opaques, now mainly leucoxene, are abundant. Chlorite with low birefringence, but apparently in part prograde, is a common accessory. Zircon was not found in a brief traverse of the thin section. A flow structure is very strong throughout the sample, but is probably late magmatic rather than metamorphic because all three major phenocryst minerals are still represented in the phenocryst assemblage. If deformation occurred in the solid state, differential fragmentation could be expected to have left one of the phases poorly represented. In particular, quartz seems to have survived with little disruption. The presence of a few subhedral quartz grains suggests that the rock crystallized as a porphyry and is not a sheared granite. GSWA Petrology Report Number 1596.

Zircon morphology

Most zircons are 75–400 µm long, colourless to dark brown, euhedral, weakly to strongly zoned, and a few have rare inclusions.

Analytical details

Heavy minerals were isolated from about 1 kg of sample using conventional heavy-liquid and magnetic techniques.

This sample was analysed on 16 and 17 March 1994. The counter deadtime was 36 ns. Thirteen analyses of the CZ-3 standard were obtained. The Pb^*/U calibration was stable with a calibration error of 2.22 (1σ%).

Results

Forty-three analyses were obtained from 40 zircons. Results are given in Table 27 and shown on a concordia plot in Figure 28.

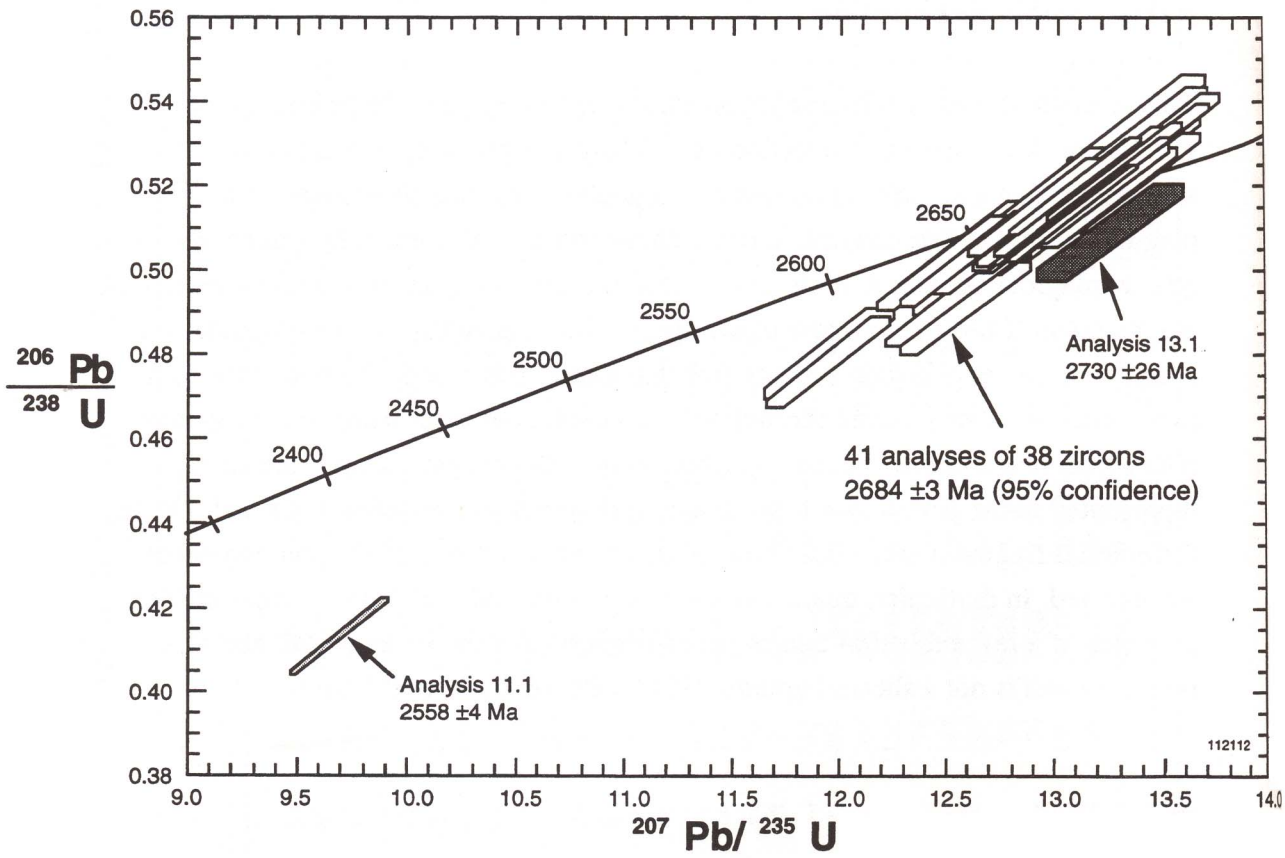

Figure 28. Concordia plot for sample 112112: K-feldspar volcanic rock, Jump Up Dam

Interpretation

Measured $^{207}Pb/^{206}Pb$ ratios of 41 analyses on 38 zircons form a single population with a little excess scatter (chi-squared = 2.50), indicating an interpreted time of emplacement of the volcanic rock at 2684 ± 3 Ma. Analysis 14.1, which has a $^{207}Pb/^{206}Pb$ age of 2730 ± 13 Ma, is interpreted to be of a xenocrystic zircon. Analysis 11.1, with a lower $^{207}Pb/^{206}Pb$ age of 2558 ± 4 Ma, has a significantly higher U concentration and is interpreted to be of a disturbed zone of a zircon originally belonging to the main population.

Table 27. Ion microprobe analytical results for sample 112112: K-feldspar volcanic rock, Jump Up Dam

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	$\pm 1\sigma$	$^{208}Pb/^{206}Pb$	$\pm 1\sigma$	$^{206}Pb/^{238}U$	$\pm 1\sigma$	$^{207}Pb/^{235}U$	$\pm 1\sigma$	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	$\pm 1\sigma$
1.1	242	144	146	0.082	0.18234	0.00053	0.16070	0.00096	0.5199	0.0116	13.07	0.30	101	2 674	5
2.1	80	36	47	0.267	0.18366	0.00100	0.11524	0.00180	0.5234	0.0119	13.25	0.32	101	2 686	9
3.1	91	33	52	0.212	0.18233	0.00102	0.09768	0.00185	0.5131	0.0116	12.90	0.31	100	2 674	9
4.1	169	94	100	0.120	0.18295	0.00065	0.14844	0.00117	0.5123	0.0115	12.92	0.30	100	2 680	6
5.1	488	342	297	0.010	0.18454	0.00034	0.19168	0.00061	0.5137	0.0114	13.07	0.29	99	2 694	3
6.1	142	78	85	0.123	0.18301	0.00071	0.14956	0.00127	0.5224	0.0117	13.18	0.31	101	2 680	6
7.1	169	88	99	0.201	0.18343	0.00070	0.13866	0.00126	0.5129	0.0115	12.97	0.30	99	2 684	6
8.1	111	49	64	0.114	0.18307	0.00077	0.11854	0.00130	0.5115	0.0115	12.91	0.30	99	2 681	7
9.1	85	29	50	0.383	0.18109	0.00113	0.08937	0.00211	0.5343	0.0121	13.34	0.32	104	2 663	10
10.1	121	42	70	1.055	0.18447	0.00130	0.10353	0.00264	0.5053	0.0114	12.85	0.31	98	2 693	12
11.1	476	259	222	0.073	0.17002	0.00037	0.13733	0.00063	0.4133	0.0092	9.69	0.22	87	2 558	4
12.1	119	47	66	0.536	0.18434	0.00106	0.10024	0.00200	0.4946	0.0111	12.57	0.30	96	2 692	9
13.1	114	60	69	0.766	0.18119	0.00109	0.14023	0.00222	0.5173	0.0117	12.92	0.31	101	2 664	10
14.1	78	29	45	0.695	0.18859	0.00152	0.10353	0.00301	0.5091	0.0116	13.24	0.33	97	2 730	13
15.1	127	58	73	0.356	0.18256	0.00085	0.12970	0.00159	0.5058	0.0114	12.73	0.30	99	2 676	8
16.1	231	91	132	1.232	0.18392	0.00089	0.11334	0.00182	0.4929	0.0110	12.50	0.29	96	2 689	8
17.1	102	32	55	0.213	0.18572	0.00092	0.08672	0.00155	0.4910	0.0111	12.57	0.30	95	2 705	8
18.1	154	59	89	1.283	0.18306	0.00111	0.09940	0.00227	0.5006	0.0112	12.64	0.30	98	2 681	10
19.1	106	47	62	1.049	0.18052	0.00135	0.11127	0.00276	0.5051	0.0114	12.57	0.31	99	2 658	12
20.1	110	37	63	0.157	0.18566	0.00095	0.09221	0.00164	0.5208	0.0118	13.33	0.32	100	2 704	8
21.1	105	37	61	0.264	0.18416	0.00111	0.09611	0.00206	0.5184	0.0117	13.16	0.32	100	2 691	10
22.1	119	58	66	1.454	0.18036	0.00145	0.10764	0.00304	0.4803	0.0108	11.94	0.30	95	2 656	13
23.1	126	47	73	0.232	0.18259	0.00090	0.10169	0.00161	0.5193	0.0117	13.08	0.31	101	2 677	8
24.1	137	70	82	0.275	0.18455	0.00087	0.13958	0.00160	0.5183	0.0117	13.19	0.31	100	2 694	8
25.1	115	43	67	0.455	0.18358	0.00100	0.09682	0.00185	0.5206	0.0118	13.18	0.31	101	2 686	9
26.1	186	104	111	0.470	0.18352	0.00081	0.15300	0.00159	0.5112	0.0115	12.93	0.30	99	2 685	7
27.1	141	65	83	0.240	0.18367	0.00083	0.12401	0.00148	0.5170	0.0116	13.09	0.31	100	2 686	7
28.1	85	35	50	0.373	0.18402	0.00115	0.11272	0.00212	0.5251	0.0119	13.32	0.32	101	2 689	10
29.1	177	91	104	0.307	0.18157	0.00073	0.13581	0.00135	0.5149	0.0116	12.89	0.30	100	2 667	7
30.1	221	87	129	0.183	0.18375	0.00059	0.10561	0.00097	0.5232	0.0117	13.26	0.30	101	2 687	5
31.1	147	53	79	0.607	0.18114	0.00097	0.10392	0.00187	0.4780	0.0107	11.94	0.28	95	2 663	9

Table 27. (*continued*)

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	$\pm 1\sigma$	$^{208}Pb/^{206}Pb$	$\pm 1\sigma$	$^{206}Pb/^{238}U$	$\pm 1\sigma$	$^{207}Pb/^{235}U$	$\pm 1\sigma$	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	$\pm 1\sigma$
32.1	100	55	61	0.237	0.18230	0.00101	0.14924	0.00188	0.5313	0.0120	13.35	0.32	103	2 674	9
33.1	307	159	182	0.268	0.18314	0.00060	0.14185	0.00111	0.5149	0.0115	13.00	0.30	100	2 682	5
34.1	195	93	112	0.606	0.18022	0.00091	0.12666	0.00177	0.5017	0.0113	12.47	0.29	99	2 655	8
1.2	401	279	245	0.099	0.18418	0.00034	0.19002	0.00061	0.5138	0.0114	13.05	0.29	99	2 691	3
1.3	266	163	162	0.186	0.18401	0.00043	0.16752	0.00078	0.5201	0.0116	13.20	0.30	100	2 689	4
35.1	212	72	121	0.982	0.18126	0.00073	0.09182	0.00146	0.5029	0.0112	12.57	0.29	99	2 664	7
36.1	95	46	58	0.393	0.18364	0.00094	0.13454	0.00183	0.5300	0.0119	13.42	0.32	102	2 686	9
37.1	124	45	72	0.146	0.18380	0.00065	0.09627	0.00103	0.5279	0.0118	13.38	0.31	102	2 687	6
38.1	142	69	85	1.036	0.18424	0.00089	0.13133	0.00182	0.5105	0.0114	12.97	0.30	99	2 691	8
39.1	109	49	65	0.887	0.18375	0.00105	0.12657	0.00213	0.5164	0.0116	13.08	0.31	100	2 687	9
40.1	94	42	56	0.381	0.18545	0.00093	0.12373	0.00174	0.5174	0.0117	13.23	0.31	99	2 702	8
7.2	113	40	64	0.320	0.18223	0.00092	0.09235	0.00165	0.5070	0.0114	12.74	0.30	99	2 673	8

STRATIGRAPHIC REFERENCE:

NELSON, D. R., 1997, Evolution of the Archaean granite–greenstone terranes of the Eastern Goldfields, Western Australia: SHRIMP U–Pb zircon constraints: Precambrian Research, v. 83, p. 57–81.

SWAGER, C. P., 1995, Geology of the greenstone terranes in the Kurnalpi–Edjudina region, southeastern Yilgarn Craton: Western Australia Geological Survey, Report 47, 31p.

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Recommended reference for this publication:

NELSON, D. R., 1995, 112112: K-feldspar volcanic rock, Jump Up Dam; in Compilation of SHRIMP U–Pb zircon geochronology data, 1994: Western Australia Geological Survey, Record 1995/3, p. 99–103.
OR
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